

HIV-1 Transmitted Drug Resistance and Genetic Diversity Among Patients From Piauí State, Northeast Brazil

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HIV-1 transmitted-drug-resistance and genetic diversity are dynamic and may differ in distinct locations/risk groups. In Brazil, increased AIDS incidence and related mortality have been detected in the Northeast region, differently from the epicenter in the Southeast. This cross-sectional study describes transmitted-drug-resistance and HIV-1 subtypes in protease/PR and reverse transcriptase/RT regions among antiretroviral naïve patients from Piauí State, Northeast Brazil. Among 96 patients recruited 89 (92.7%) had HIV-1 PR/RT regions sequenced: 44 females and 45 males, 22 self-declared as men who have sex with men. Transmitted-drug-resistance was investigated by CPR tool (Stanford HIV-1 Drug Resistance/SDRM). HIV-1 subtypes were assigned by REGA and phylogenetic inference. Overall, transmitted-drug-resistance rate was 11.2% (10/89; CI 95%: 5.8–19.1%); 22.7% among men who have sex with men (5/22; CI 95%: 8.8–43.4%), 10% in heterosexual men (2/20; CI 95%: 1.7–29.3%) and 6.8% in women (3/44; CI 95%: 1.8–17.4%). Singleton mutations to protease-inhibitor/PI, nucleoside-reverse-transcriptase-inhibitor/NRTI or non-nucleoside-reverse-transcriptase-inhibitor/NNRTI predominated (8/10): PI mutations (M46L, V82F, L90M); NRTI mutations (M41L, D67N) and NNRTI mutations (K103N/S). Dual class resistance mutations to NRTI and NNRTI were observed: T215L (NRTI), Y188L (NNRTI) and T215N (NRTI), F227L (NNRTI). Subtype B prevailed (86.6%; 77/89), followed by subtype F1 (1.1%, 1/89) and subtype C (1.1%, 1/89). B/F1 and B/C intersubtype recombinants represented 11.2% (10/89). In Piauí State extensive testing of incidence and transmitted-drug-resistance in all populations with risk behaviors may help control AIDS epidemic locally. **J. Med. Virol. 87:798–806, 2015.**

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KEY WORDS: HIV-1; transmitted drug resistance; subtypes; Northeast Brazil

INTRODUCTION

HIV-1 surveillance programs should monitor incidence/prevalence, subtypes distribution and drug resistance mutations among all exposure categories, especially in high risk groups, such as men who have sex with men, commercial sex workers and intravenous drug users (IDU). Brazil has reported the largest number of AIDS cases and the highest level of HIV-1 genetic diversity in Latin America and in this huge and diverse country, AIDS epidemic reflects important regional differences [Brazil, 2013]. Although the epidemic has been considered stable, over the last 10 years, from 2003 to 2012 an

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This study was approved by the regional Institutional Review Boards (“Comitê de Ética em Pesquisa da Faculdade de Ciências Médicas da Universidade Estadual do Piauí”, protocol# 022/2011 and “Comitê de Ética de Pesquisa do Hospital de Doenças Tropicais Dr. Natan Portela”, protocol# 16/2011). All participants were instructed about the study and signed a written informed consent before interview and blood collection for HIV-1 molecular analysis.

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important increase in AIDS detection rate (62.6%) has been observed in the Northeast region of the country [Brazil, 2013].

Successful antiretroviral (ARV) treatment, which comprises combinations of nucleoside and non-nucleoside reverse transcriptase inhibitors (NRTIs/NNRTIs) and protease inhibitors (PIs), is important to reduce viral load and virus transmission [Shafer, 2002; Shafer and Schapiro, 2008]. Distribution of ARV drugs by public health programs has been scaled up, raising concerns about the selection of viral mutants associated with drug resistance, as they can lead to virological failure, secondary resistance and transmission of resistant strains [Shafer, 2002; Bennett et al., 2008; Wainberg et al., 2011]. Transmitted drug resistance (TDR) is an important public health issue that can jeopardize reductions in mortality, morbidity, and transmission [Pineda-Peña et al., 2012; WHO, 2012].

The Brazilian AIDS epidemic is highly concentrated in the Southeast and South regions and several studies from these regions have reported TDR levels in different study groups [Inocencio et al., 2009; Sprinz et al., 2009; Sanabani et al., 2011; Ferreira et al., 2013; Pilotto et al., 2013; UNAIDS, 2013]. However, there is scanty data on the HIV-1 molecular epidemiology in other regions away from the epicenter of the epidemics. This is especially true for the Northeast region that accounts for 88,830 reported AIDS cases with previous reports from only three out of its nine states (Pernambuco, Ceará, and Bahia States) [de Medeiros et al., 2006; Pedroso et al., 2007; Arruda et al., 2011; Cavalcanti et al., 2012]. Overall in Brazil, low to moderate levels of TDR have been described, and recommendation of pre-ARV therapy genotyping is restricted to pregnant women and to those infected by patients under ARV therapy.

Aiming to reduce the gap in information about molecular characteristics of the AIDS epidemic in Northeast Brazil, this study describes HIV-1 TDR and subtypes circulating among ARV naïve patients, heterosexual and men who have sex with men living in Piauí State. Recent official data show 133.9% increase in AIDS detection rate in Piauí State from 2003 to 2012 [Brazil, 2013]. Data on the molecular epidemiology of HIV-1 from a Brazilian region of increasing incidence of AIDS cases can contribute to better preventive measures.

MATERIALS AND METHODS

Study Area and Subjects

This is a descriptive cross-sectional study designed to investigate the molecular epidemiology of HIV-1 among ARV naïve patients attending public health services in Piauí State, Northeast Brazil (3,118,360 inhabitants; Teresina city, the capital: 814,230 inhabitants). Piauí State is characterized by small agriculture-centered cities (soybeans, rice, corn, and cotton) and livestock production farms. Approximately half of

patients included in this survey lived in Teresina and the remaining ones came from 28 different small villages located throughout the State.

In Piauí State, 3,958 AIDS cases were reported from 2009 to 2013 and during the study period (August 2011 to May 2012) 96 consecutive cases of HIV-1 ARV naïve patients were recruited. Patients were recruited in Teresina while attending the Public Health Central Laboratory, which is the reference health service responsible for performing CD4⁺ T cell counts and viral load determinations. The study group represented 28.7% of all ARV naïve patients (n = 334) that attended this reference center during the study period. Inclusion criteria were: patients of both genders, older than 18 years, with recent or previous diagnosis of HIV-1 infection, having no previous history of ARV exposure and living in Piauí State. Standardized questionnaires were used by a trained person to obtain socio-behavioral data by face-to-face interviews. Clinical and laboratory data, including plasma viral load (bDNA; Siemens Medical Solutions, Mississauga, Canada) and CD4⁺ T cell counts (FACSalibur, Becton & Dickson, San Jose, CA) were retrieved from medical files at a reference hospital (Hospital de Doenças Tropicais, Dr. Natan Portela) in Teresina city.

Genetic Analysis

HIV-1 RNA was extracted from plasma (QIAamp Viral RNA Mini Kit, Qiagen, Hilden, Germany), reverse transcribed into complementary DNA (cDNA) (Invitrogen, Carlsbad, CA) and used as the target for a nested polymerase chain reaction (nested-PCR) of *pol* region as previously described [Cardoso et al., 2009]. The entire protease (PR) fragment and approximately 750-bp of the reverse transcriptase (RT) fragment were amplified [Frenkel et al., 1995; Kozal et al., 1996]. After purification of amplicons (QIAquick PCR Purification Kit/Qiagen, Hilden, Germany), genomic sequencing was performed (ABI Prism 3130 Genetic Analyzer, Applied Biosystems, Foster City, CA).

Phylogenetic Analysis

HIV-1 subtypes were identified by REGA tool version 2.0 and by phylogenetic inference using reference sequences obtained from Los Alamos HIV database [Oliveira et al., 2005]. Trees were generated by Neighbor-Joining method under Kimura's two-parameter model using MEGA5 software [Kimura, 1980; Tamura et al., 2011]. Bootstrap values (1,000 replicates) above 70% were considered significant. The HIV-1 sequences with discordant subtypes in the PR and RT fragments, which are suggestive of intersubtype recombination, were analyzed by SIM-PLLOT 3.5.1 [Lole et al., 1999].

GenBank accession numbers of reference sequences used in the comparative phylogenetic and SIM-PLLOT analyses are the following, for subtype B: U21135,

FJ638434, AY173956, K03455, EF379194, EF379210, FJ548795, FJ548804, and U63632; for subtype C: AF110967, U46016, FJ548791, FJ594149, AF286228, AF067155, and U52953; for subtype F1: DQ358801, AF077336, and AF005494; and for the simian immunodeficiency virus sequence from the chimpanzee (SIVcpz) used as an outgroup: X52154. GenBank accession numbers of HIV-1 *pol* sequences described in this study are KF700948-KF701036.

Resistance Analysis

TDR rate was analyzed using the Calibrated Population Resistance (CPR) tool (Stanford Surveillance Drug Resistance Mutation-SDRM) [Gifford et al., 2009]. Susceptibility profile of ARV mutations was analyzed by the SDRM-2009 (Accessed: October/2013).

Statistical Analysis

Continuous variables were described as medians with minimum and maximum range. Categorical variables were presented as frequency and percentage. Chi-square or Fishers exact test were used to analyze the differences among categorical variables, with a significance level of 5%. All descriptive analyses were performed using Epi Info™ version 7 (CDC, Atlanta).

RESULTS

Among 96 ARV naïve patients recruited, 89 had both the PR and RT regions of HIV-1 sequenced (89/96, 92.7%). The median age of participants was 34

years (range: 18–80 years), 45 were males and among them, 22 self-reported as men who have sex with men. Heterosexual unprotected sex was reported by 64 patients and one participant was an IDU. The median of CD4⁺ T cell counts was 424 cells/μl (range: 27–1,290 cells/μl) and most patients (78/89) had HIV-1 plasma viral loads below 100,000 copies/ml (Table I). Baseline and virologic characteristics of study subjects are described in Table I.

Overall, the rate of TDR among patients from Piauí State was 11.2% (10/89; CI 95%: 5.8–19.1%). Approximately half (5/12) of HIV-1 isolates with resistance mutations were isolated from men who have sex with men (Table I). The TDR rate among men who have sex with men was 22.7% (5/22; CI 95%: 8.8–43.4%) and among heterosexual males it was 10% (2/20; CI 95%: 1.7–29.3%). Among women, TDR rate was 6.8% (3/44; CI 95%: 1.8–17.4%). However these differences did not reach statistical significance ($P=0.159$). Patients with TDR ($n=10$) had a median of CD4⁺ T cell counts of 380 cells/μl (range 27–678 cells/μl); six patients had viral loads between 10,000–100,000 copies/ml. Although patients from this study came from 28 cities located throughout Piauí State, TDR cases were concentrated mainly in the North and Central regions, near the capital, Teresina (Fig. 1).

Mutation profiles of ARV resistant HIV-1 isolates and the level of resistance to distinct ARV drugs are depicted in Table II. Most isolates (10/12) had single class resistance mutations: PI mutations (M46L, V82F, L90M); NRTI mutations (M41L, D67N); NNRTI mutations (K103N/S, E138A). Dual class resistance mutations to NRTI and NNRTI were

TABLE I. Baseline and Virologic Characteristics of 96 HIV-1 Antiretroviral Naïve Subjects From Piauí State, Northeast Brazil

	ARV naïve patients (n = 89)	Transmitted drug resistance (SDRM-2009) (n = 10/89, 11.2%)
Gender		
Male	45 (50.6%)	7 (15.5%)
Female	44 (49.4%)	3 (6.8%)
Median of age in years (range)	34 (18–80)	36 (27–48)
Route of infection [n (%)]		
Heterosexual	64 (71.9%)	6 (9.4%)
Female	44 (49.4%)	3 (6.8%)
Male	20 (22.5%)	2 (10.0%)
MSM	22 (24.7%)	5 (22.7%)
IDU	1 (1.1%)	1 (100.0%)
Unknown	2 (2.2%)	—
Median of CD4 counts, cells/μl (range)	424 (27–1,290)	380 (27–678)
HIV-1 RNA, copies/ml [n (%)]		
<10,000	39 (43.8%)	4 (33.3%)
10,000–100,000	39 (43.8%)	6 (50.0%)
>100,000	11 (12.4%)	2 (16.7%)
PR/RT HIV-1 subtype [n (%)]		
B/B	77 (86.5%)	9 (90.0%)
F1/F1	1 (1.1%)	—
C/C	1 (1.1%)	—
B/F1	4 (4.5%)	—
B/C	6 (6.7%)	1 (10.0%)

ARV, antiretroviral; IDU, intravenous drug user; MSM, men who have sex with men; SDRM, Stanford surveillance drug resistance mutation; PR, protease; RT, reverse transcriptase.

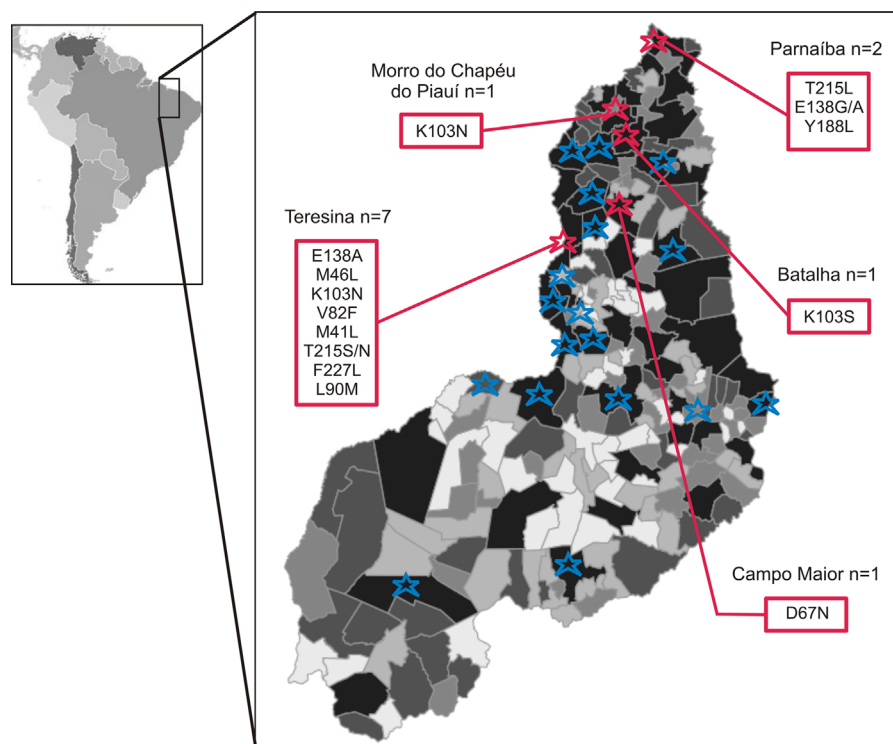


Fig. 1. South America map highlighting the study area-Piauí State and its capital, Teresina city, in Northeast Brazil. Municipalities are classified by population density: dark gray marks represent cities densely populated (11,000–815,000 inhabitants), light gray marks represent less populated cities (1,000–4,500 inhabitants). Stars without descriptions represent the place of residence of HIV-1 infected subjects; Highlighted stars represent municipalities where subjects with transmitted drug resistance were identified.

observed among two isolates: first one (T215L-NRTI and E138G, Y188L-NNRTI), second one (T215N-NRTI and F227L-NNRTI).

A clear predominance of sequences clustering with a single subtype was observed (88.8%, 79/89) (Fig. 2). Among isolates with concordant subtypes in PR and RT regions, subtype B predominated (86.6%; 77/89). Non-subtype B isolates were rare: only one patient harbored subtype F1 (1.1%; 1/89) and one case of subtype C infection was identified (1.1%; 1/89). Moreover, B/F1 and B/C recombinant forms were identified in 10 female patients (11.2%; 10/89): F1^{PR}/B^{RT} (n = 1), F1^{PR}/BF1B^{RT} (n = 1), B^{PR}/BF1^{RT} (n = 2), B^{PR}/BCB^{RT} (n = 2), B^{PR}/C^{RT} (n = 1), B^{PR}/BC^{RT} (n = 1), C^{PR}/CB^{RT} (n = 1) and one CRF31_BC-like isolate. Recombination patterns of the mosaics are presented in Figure 3. Among 12 isolates with resistance mutation, 11 were classified as subtype B and one was a B^{PR}/BC^{RT} recombinant form.

DISCUSSION

In this study, moderate rate of TDR was observed in Piauí State, Northeast Brazil, where incidence of both AIDS cases and related mortality has increased significantly in the last decade. Men who have sex with men presented a higher trend of TDR (>20%)

compared to heterosexual male and female patients (~10%). However, TDR rate between men who have sex with men and heterosexual patients was not statistically significant. Other studies including men who have sex with men have shown moderate to high prevalence of resistance mutations [Cardoso et al., 2009; Bermudez-Aza et al., 2011; Karlsson et al., 2012; Tupinambas et al., 2013]. Despite the increased risk for HIV infection among men who have sex with men, extensive testing of incidence and TDR level should be applied to all populations with risk behaviors and not only men who have sex with men.

Increased concern about resurging HIV-1 incidence among men who have sex with men had arisen on a global level, with high incidence rates reported in Asia, Africa, Latin America, and the former Soviet Union [Bautista et al., 2004; Wade et al., 2005; Baral et al., 2007; van Griensven, 2007; van Griensven et al., 2009]. In Brazil, the prevalence of HIV-1 infection in this population group raised from 0.56% in 2002 to 1.2% in 2007 [Szwarcwald et al., 2011]. In the current study, almost half of patients with TDR were men who have sex with men and some of them (5 out of 22) reported bisexual behavior, confirming their potential bridging role in the transmission of HIV-1 infection to heterosexual women. The reasons for the apparently higher trend of TDR among this

TABLE II. Amino Acid Substitutions in Protease and Reverse Transcriptase Resistance-Related Codons of HIV-1 Isolates From ARV Naïve Patients With Transmitted Drug Resistance From Piauí State, Northeast Brazil

GenBank accession number	Route of infection HIV-1 PR/RT subtype	Sex/age (years)	Mutations				Resistance profile			
			PI minor	PI major	NRTI	NNRTI	Low	Intermediate	High	
KF700966	Hetero B	F/34	—	M46L	—	—	NFV	—	—	—
KF700975	Hetero B/BC	F/46	—	—	—	K103N	—	—	—	EFV, NVP
KF700988	Hetero B	M/43	—	V82F	—	—	ATV, DRV, LPV	FPV, IDV, NFV	—	—
KF700993	MSM B	M/35	—	—	D67N, V75I	—	AZT, d4T	—	—	—
KF700995	Hetero B	F/39	—	—	M41L, T215S	—	ABC, ddI, TDF	AZT, d4T	—	—
KF700996	MSM B	M/31	—	—	—	K103N	—	—	—	EFV, NVP
KF701005	MSM B	M/48	—	—	—	K103S	—	EFV	—	NVP
KF701010	MSM B	M/27	L10V	—	T215L	V179E, Y188L	AZT, d4T	ETR	—	EFV, NVP
KF701014	MSM B	M/43	—	—	T215N	F227L	AZT, d4T, EFV	NVP	—	—
KF701030	Hetero B	M/35	L10V	L90M	—	—	ATV, FPV, IDV	SQV	—	NFV

PR, protease region; RT, reverse transcriptase region; MSM, men who have sex with men; Hetero, heterosexual; M, Male; F, Female; PI, protease inhibitor; NRTI, nucleoside reverse transcriptase inhibitors; NNRTI, non-nucleoside reverse transcriptase inhibitors, ATV, atazanavir; AZT, zidovudine; ddI, didanosine; d4T, stavudine; DRV, darunavir; EFV, efavirenz; FPV, fosamprenavir; IDV, indinavir; LPV, lopinavir; NFV, nelfinavir; NVP, nevirapine; SQV, saquinavir; TDF, tenofovir. The resistance mutations are given in bold.

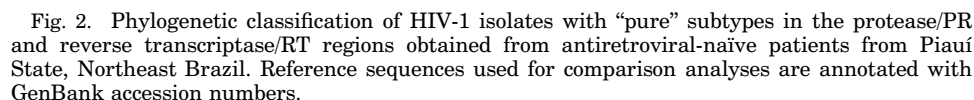
group from Piauí are unclear and do not reflect time since diagnosis as four out of five men who have sex with men with TDR were not considered recently diagnosed or recently infected cases as their date of diagnosis was before 2012. The longer the time since diagnosis, the fewer the chances to detect transmitted resistance mutations.

The small number of men who have sex with men in the studied population, limits further generalization. However, representative samples of high risk subpopulations are a well-known surveillance challenge. High risk groups have been usually under-represented in surveillance evaluations and in targeted prevention programs. Men who have sex with men can still be considered a “hidden” group as they may prefer not to participate in surveillance studies fearing stigma and discrimination [Heckathorn, 2002]. Local prevention and intervention programs on high risk populations may be important to interrupt transmission of HIV-1 and TDR.

The overall moderate rate of TDR identified in the current study (11.2%) is in accordance with reports from other geographical regions in Brazil: 11.5% in the Northern Tocantins State; 10% in the Central Western Goiás State; 14.2% in the Southeastern São Paulo State, and 11% in the Southern Santa Catarina State [Cardoso et al., 2009; Carvalho et al., 2011; Graf et al., 2011; Ferreira et al., 2013]. Two recent multicenter surveys in Brazil have shown moderate levels of TDR [Inocencio et al., 2009; Sprinz et al., 2009].

In ARV naïve patients from Piauí State, singleton mutations to PI, NRTI or NNRTI predominated (8/10); only two patients had dual-class resistance mutations (NRTI/NNRTI): T215L (NRTI) and Y188L (NNRTI); T215N (NRTI) and F227L (NNRTI). The most frequent NNRTI resistance mutations were: K103N/S (30%; 3/10) that causes high-level resistance to nevirapine and efavirenz [Bennett et al., 2009]. Major PI resistance mutations (M46L, V82F, and L90M) were detected. A previous study among patients under PI-based highly active antiretroviral therapy (HAART) from the United States of America showed L90M as the most prevalent (42%) resistance mutation [Kantor et al., 2002].

Thymidine analog mutations (TAMs) are frequent in low-income countries with ARV treatment based on fixed-dose combinations of thymidine analogs. In this study, type I TAM (M41L) and type II TAM (D67N) were observed. The M41L mutation occurred with T215S but M41L usually occurs with T215Y/F mutations, causing high-level resistance to zidovudine, stavudine and intermediate resistance to didanosine, abacavir, and tenofovir [Marcelin et al., 2004]. The T215S/N/L mutation was the most frequent NRTI, identified in 3 out of 10 patients, two of them were men who have sex with men. T215S/N is a revertant mutation that indicates that the T215Y/F mutation was previously transmitted to one zidovudine-naïve patient and that the virus reverted to a



The AIDS epidemic in Piauí State is characterized by low level of HIV-1 genetic diversity: more than 85% of the patients were infected with “pure” subtype B. This finding contrasts with the declining trend in subtype B prevalence reported in other Brazilian regions such as in the south, southeast, and central western [Cardoso et al., 2009; Cardoso et al., 2011; Ferreira et al., 2011; da Silveira et al., 2012]. Teresina city is the only capital in the Northeast region which is not located by the Atlantic Coast and therefore has limited touristic attractions, a factor that may have contributed to the highly prevalent subtype-B epidemic. In this study group, only two

These results indicate low northwards dissemination of “pure” subtype C, which is prevalent in southern Brazil [Vicente et al., 2000; de Medeiros et al., 2006; Cavalcanti et al., 2007; Pedrosa et al., 2007; Cardoso et al., 2011; Cavalcanti et al., 2012; Delatorre et al., 2013]. Diverse B/F1 and B/C recombinants representing 11.2% of the viruses were found where circulating “pure” subtype F1 and C isolates were rare. In this context, there are reduced chances of subtypes B/F1 or B/C co-infection or super-infection which is required for the generation of recombinants. Therefore, rather than being generated locally, the B/F1 and B/C recombinant forms circulating in Piauí have probably been imported from other Brazilian regions where they are frequent. The results from this study also show that subtype C seems to be spreading to Northeast Brazil as B/C recombinants, not as “pure” subtype C viruses. Full-length or near full-length genomes of B/F1 and B/C recombinant viruses can indicate whether they represent any of the already described CRFs in Brazil or elsewhere or if they represent new yet unidentified CRFs.

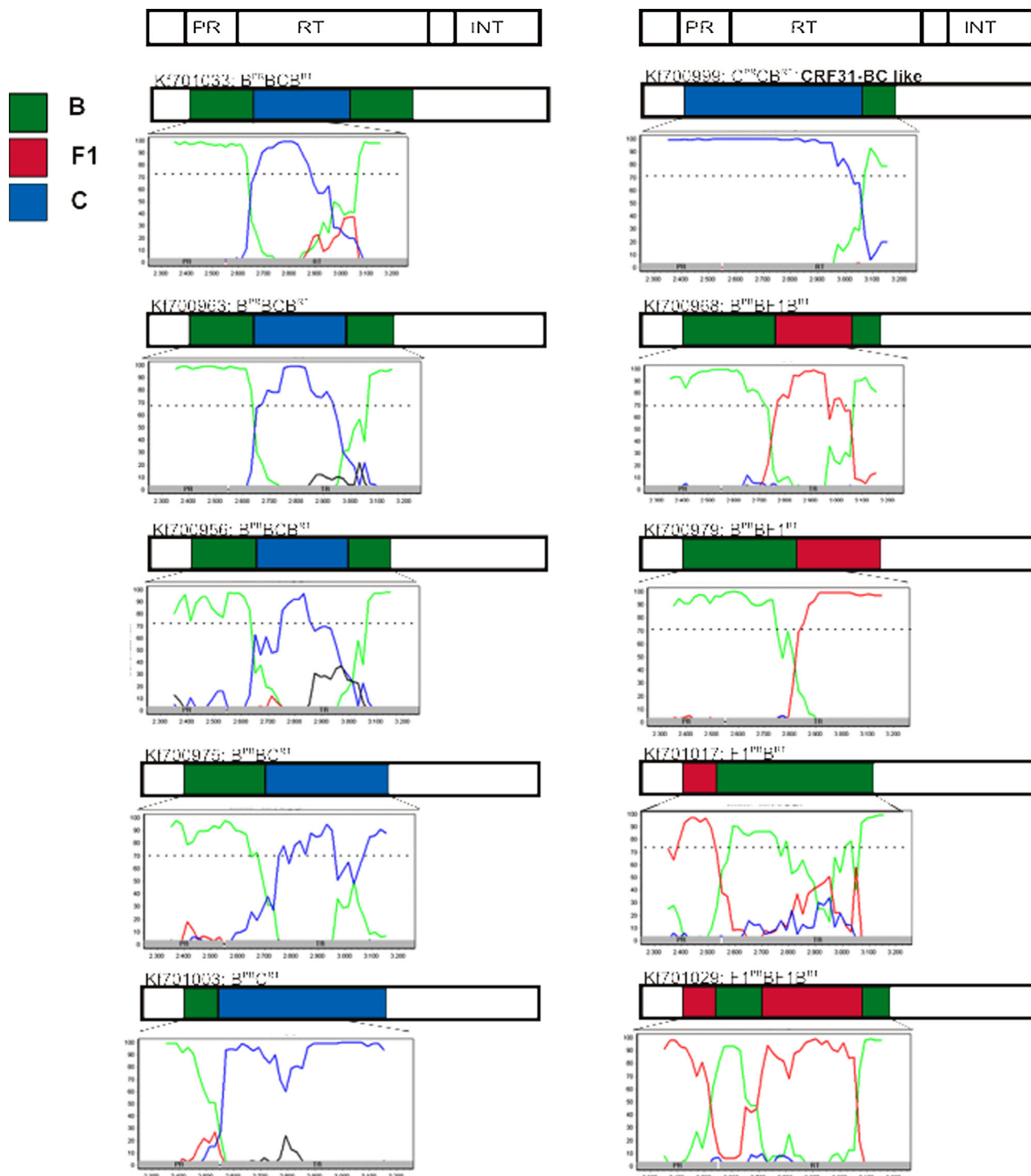


Fig. 3. Bootscanning analysis of BF1 and BC recombinant HIV-1 sequences in protease and reverse transcriptase regions from 10 ARV-naïve patients from Piauí State, Northeast Brazil. Bars represent HIV-1 PR, protease; RT, reverse transcriptase; INT, integrase fragments in pol gene, and the different HIV-1 subtypes.

In conclusion, in this study the high subtype B prevalence and the limited circulation of non-subtype B viruses, contrast with several patterns of B/F1 and B/C recombinants. Moderate rate of TDR in this area

of increasing HIV-1 prevalence, highlights the need for surveillance studies to monitor TDR level, which is considered the cornerstone for pre ARV genotyping guidelines.

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